

age	income	student	credit_rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$\text{Info}(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.409 - (-0.531) = 0.940$$

$$\begin{aligned} \text{Info}_{\text{age}}(D) &= \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) \\ &= \frac{5}{14} \left(-\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \right) + \frac{4}{14} \left(-\frac{4}{4} \log_2\left(\frac{4}{4}\right) - \frac{0}{4} \log_2\left(\frac{0}{4}\right) \right) + \frac{5}{14} \left(-\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \right) \\ &= 0.347 + 0 + 0.347 \\ &= 0.694 \end{aligned}$$

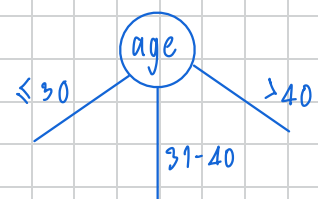
$$\begin{aligned} \text{Info}_{\text{income}}(D) &= \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2) \\ &= \frac{4}{14} \left(-\frac{3}{4} \log_2\left(\frac{3}{4}\right) - \frac{1}{4} \log_2\left(\frac{1}{4}\right) \right) + \frac{6}{14} \left(-\frac{4}{6} \log_2\left(\frac{4}{6}\right) - \frac{2}{6} \log_2\left(\frac{2}{6}\right) \right) + \frac{4}{14} \left(-\frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right) \right) \\ &= 0.232 + 0.393 + 0.286 \\ &= 0.911 \end{aligned}$$

$$\begin{aligned} \text{Info}_{\text{student}}(D) &= \frac{7}{14} I(6,1) + \frac{7}{14} I(3,4) \\ &= \frac{7}{14} \left(-\frac{6}{7} \log_2\left(\frac{6}{7}\right) - \frac{1}{7} \log_2\left(\frac{1}{7}\right) \right) + \frac{7}{14} \left(-\frac{3}{7} \log_2\left(\frac{3}{7}\right) - \frac{4}{7} \log_2\left(\frac{4}{7}\right) \right) \\ &= 0.296 + 0.493 \\ &= 0.789 \end{aligned}$$

$$\begin{aligned} \text{Info}_{\text{credit_rating}}(D) &= \frac{6}{14} I(3,3) + \frac{8}{14} I(6,2) \\ &= \frac{6}{14} \left(-\frac{3}{6} \log_2\left(\frac{3}{6}\right) - \frac{3}{6} \log_2\left(\frac{3}{6}\right) \right) + \frac{8}{14} \left(-\frac{6}{8} \log_2\left(\frac{6}{8}\right) - \frac{2}{8} \log_2\left(\frac{2}{8}\right) \right) \\ &= 0.429 + 0.463 \\ &= 0.892 \end{aligned}$$

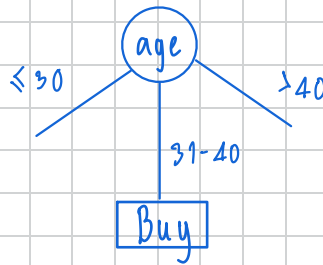
เลือก age

$$\begin{aligned} \text{Gain}(\text{age}) &= \text{Info}(D) - \text{Info}_{\text{age}}(D) = 0.940 - 0.694 = 0.246 \\ \text{Gain}(\text{income}) &= \text{Info}(D) - \text{Info}_{\text{income}}(D) = 0.940 - 0.911 = 0.029 \\ \text{Gain}(\text{student}) &= \text{Info}(D) - \text{Info}_{\text{student}}(D) = 0.940 - 0.789 = 0.151 \\ \text{Gain}(\text{credit_rating}) &= \text{Info}(D) - \text{Info}_{\text{credit_rating}}(D) = 0.940 - 0.892 = 0.048 \end{aligned}$$



age > student > credit_rating > income

age	student	buys computer
<=30	no	no
<=30	no	no
31...40	no	yes
...	...	...
31...40	yes	yes
<=30	no	no
<=30	yes	yes
...	...	...
<=30	yes	yes
31...40	no	yes
31...40	yes	yes



Buy	Age		
	<=30	31-40	>40
yes	2	4	3
no	3	0	2

จำนวน 31-40 ซื้อแน่นอน

พิจารณาแค่ ≤ 30 และ > 40

$$\text{Info}(D) = I(5,5) = -\frac{5}{10} \log_2 \left(\frac{5}{10} \right) - \frac{5}{10} \log_2 \left(\frac{5}{10} \right) = 1$$

$$\begin{aligned} \text{Info}_{\leq 30}(D) &= \frac{2}{5} I(2,0) + \frac{3}{5} I(0,3) \\ &= \frac{2}{5} \left(-\frac{2}{2} \log_2 \left(\frac{2}{2} \right) - \frac{0}{2} \log_2 \left(\frac{0}{2} \right) \right) + \frac{3}{5} \left(-\frac{0}{3} \log_2 \left(\frac{0}{3} \right) - \frac{3}{3} \log_2 \left(\frac{3}{3} \right) \right) \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Info}_{>40}(D) &= \frac{3}{5} I(2,1) + \frac{2}{5} I(1,1) \\ &= \frac{3}{5} \left(-\frac{2}{3} \log_2 \left(\frac{2}{3} \right) - \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right) + \frac{2}{5} \left(-\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \\ &= 0.551 + 0.4 \\ &= 0.951 \end{aligned}$$

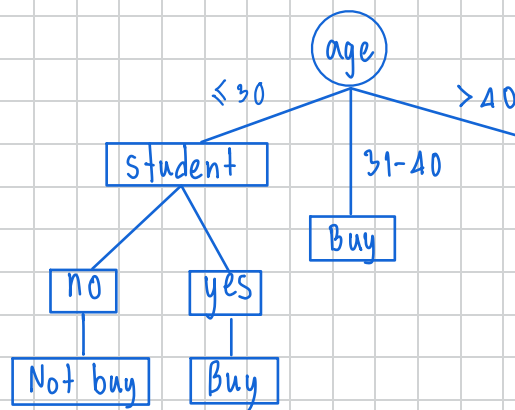
$$\text{Gain}(\leq 30) = \text{Info}(D) - \text{Info}_{\leq 30}(D) = 1 - 0 = 1$$

$$\text{Gain}(>40) = \text{Info}(D) - \text{Info}_{>40}(D) = 1 - 0.951 = 0.049$$

เลือก ≤ 30

Buy	Student	
	yes	no
yes	3	2
no	0	3

คนที่ เป็น นักเรียน
ซื้อแน่นอน



เหลือ 2 ตัวแปร
credit_rating / income?

age	income	credit_rating	buys_computer
>40	medium	fair	yes
>40	low	fair	yes
>40	low	excellent	no
>40	medium	fair	yes
>40	medium	excellent	no

$$\text{Info}(D) = I(3,2) = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) = 0.442 - (-0.519) = 0.971$$

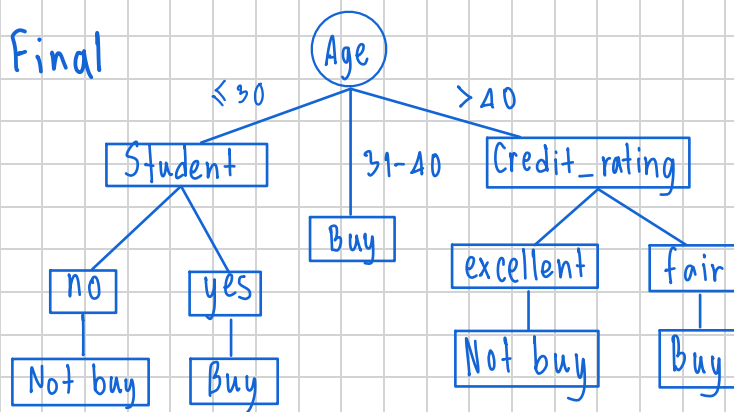
$$\begin{aligned} \text{Info}_{\text{income}}(D) &= \frac{3}{5} I(2,1) + \frac{2}{5} I(1,1) \\ &= \frac{3}{5} \left(-\frac{2}{3} \log_2 \left(\frac{2}{3} \right) - \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right) + \frac{2}{5} \left(-\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \\ &= 0.551 + 0.4 \\ &= 0.951 \end{aligned}$$

$$\begin{aligned} \text{Info}_{\text{credit_rating}}(D) &= \frac{2}{5} I(0,2) + \frac{3}{5} I(3,0) \\ &= \frac{2}{5} \left(-\frac{0}{2} \log_2 \left(\frac{0}{2} \right) - \frac{2}{2} \log_2 \left(\frac{2}{2} \right) \right) + \frac{3}{5} \left(-\frac{3}{3} \log_2 \left(\frac{3}{3} \right) - \frac{0}{3} \log_2 \left(\frac{0}{3} \right) \right) \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

$$\text{Gain}(\text{income}) = \text{Info}(D) - \text{Info}_{\text{income}}(D) = 0.971 - 0.951 = 0.020$$

$$\text{Gain}(\text{credit_rating}) = \text{Info}(D) - \text{Info}_{\text{credit_rating}}(D) = 0.971 - 0 = 0.971$$

เลือก credit_rating



Age > 40

Buy	credit_rating	
	excellent	fair
yes	0	3
no	2	0

excellent → ไม่ซื้อ
fair → ซื้อ